Frozen Lake Game using Q-Learning

Machine Learning Lab Project

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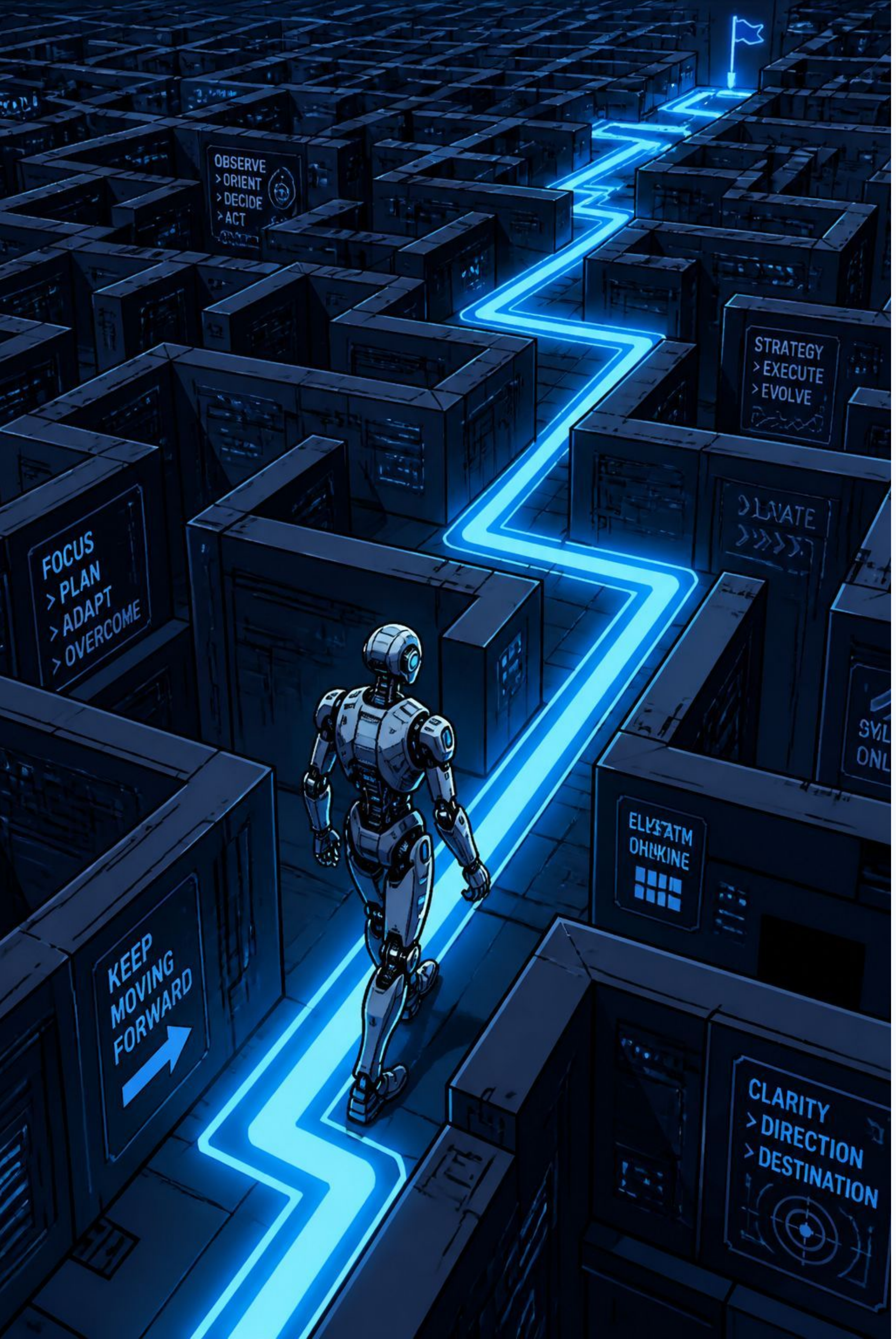


Introduction & Objective

Introduction

Reinforcement Learning (RL) is a branch of Machine Learning where an agent learns by interacting with an environment. The agent receives rewards for correct actions and penalties for incorrect actions. Over time, it learns the best strategy to maximize rewards.

Objective

- Train an intelligent agent using the Q-Learning algorithm.
- Navigate the Frozen Lake safely.
- Reach the goal while avoiding holes.
- Compare the trained agent with a random agent.
- Evaluate the learning performance using graphs and success rate.

Technologies & Methodology

Technologies Used

Python

Gymnasium

FrozenLake-v1

NumPy

Matplotlib

Pandas

Google Colab

Methodology

01

Create the FrozenLake environment.

02

Initialize the Q-table with zeros.

03

Train the agent using the Q-Learning algorithm.

04

Update Q-values using rewards received from the environment.

05

Reduce exploration gradually using epsilon decay.

06

Test the trained agent and compare it with a random agent.

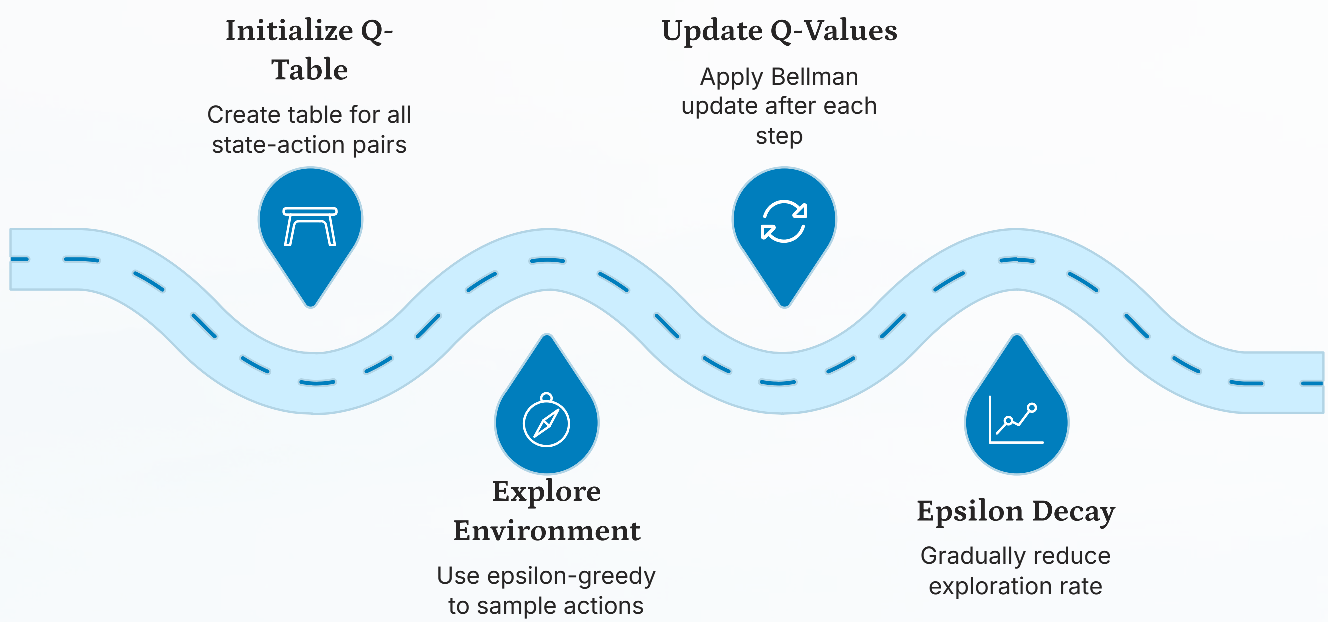
Implementation & Results

Implementation

- Created a 4×4 FrozenLake environment.
- Initialized a Q-table for all state-action pairs.
- Trained the agent for 10,000 episodes.
- Used the epsilon-greedy strategy for exploration and exploitation.
- Stored rewards after every episode.

Results

- The trained agent learned the optimal path to the goal.
- The success rate improved significantly after training.
- The reward graph showed steady learning progress.
- The trained agent performed much better than the random agent.



The agent iterates through 10,000 episodes, progressively shifting from exploration to exploitation as epsilon decays, converging on the optimal path.

Advantages & Applications

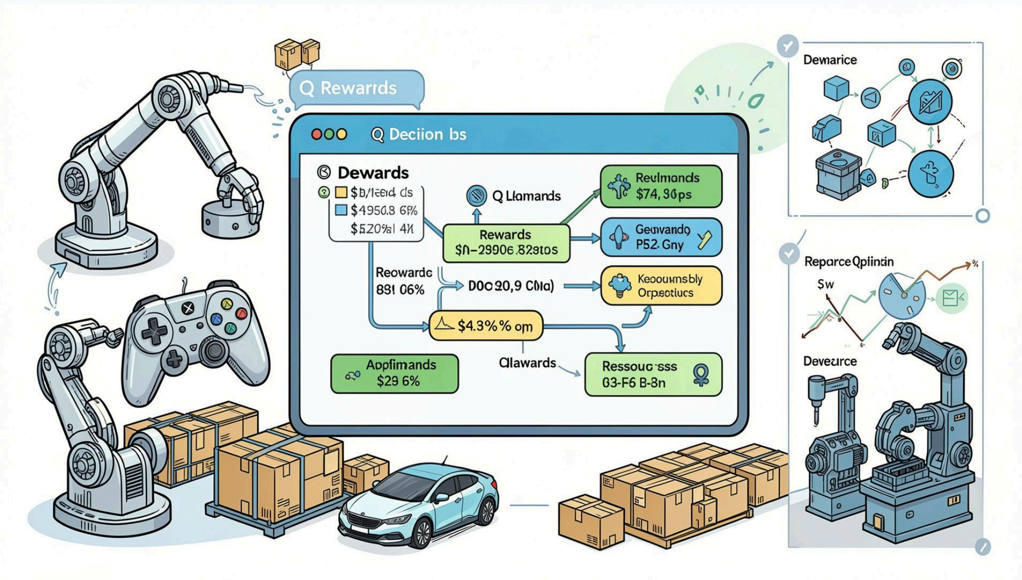
Advantages

Learns through trial and error.

Does not require labeled data.

Improves decision-making over time.

Can adapt to different environments.



Applications

Robotics

Game AI

Self-driving Vehicles

Warehouse Automation

Industrial Optimization

Resource Management

Conclusion & Future Scope

Conclusion

This project successfully demonstrated the Q-Learning algorithm using the FrozenLake environment. The agent learned the optimal path through continuous interaction with the environment. Compared with a random agent, the trained agent achieved a much higher success rate, proving the effectiveness of Reinforcement Learning.

Future Scope

Deep Q-Network (DQN)

Implement DQN for more complex decision-making.

Larger Environments

Train on larger and more complex environments.

Real-World Robotics

Apply RL to real-world robotic navigation tasks.

Advanced RL Algorithms

Improve learning speed using advanced RL methods.